A **July 30, 2026** SOPHIE CHEN

Launching Off a Planet [12 pts]

The gravitational parameter is

$$GM = (6.67 \times 10^{-11})(2.16 \times 10^{24}) = 1.44 \times 10^{14}\ \mathrm{m^3\,s^{-2}}.$$

Q1. Orbital speed

The orbital radius is $r = R + h = 6.025 \times 10^6$ m. For a circular orbit the gravitational force supplies the centripetal force, so

$$\frac{GMm}{r^2} = \frac{mv_{\mathrm{orbit}}^2}{r} \implies v_{\mathrm{orbit}} = \sqrt{\frac{GM}{r}} = \sqrt{\frac{1.44 \times 10^{14}}{6.025 \times 10^6}} = \boxed{4.89 \times 10^3\ \mathrm{m/s}}.$$

Q2. Vertical launch speed

The surface gravity is

$$g = \frac{GM}{R^2} = \frac{1.44 \times 10^{14}}{(6 \times 10^6)^2} = 4.00\ \mathrm{m/s^2}.$$

In the flat, uniform-field approximation the traveller reaches the orbit altitude h when his vertical velocity has been reduced to zero (the top of the ballistic arc, where he is moving purely tangentially). With $v_y^2 = 2gh$,

$$\boxed{v_y = \sqrt{2gh} = \sqrt{2(4.00)(2.5 \times 10^4)} = 447\ \mathrm{m/s}}.$$

Q3. Initial velocity

The tangential component is $v_x = v_{\mathrm{orbit}}$ (no horizontal force in the flat approximation), so

$$v_0 = \sqrt{v_x^2 + v_y^2} = \sqrt{(4.89 \times 10^3)^2 + (447)^2} = \boxed{4.91 \times 10^3\ \mathrm{m/s}},$$

at an angle

$$\theta = \arctan\frac{v_y}{v_x} = \arctan\frac{447}{4890} = \boxed{5.2^\circ}$$

above the tangent.

Q4. Spring compression

The spring stores elastic energy $\frac{1}{2}kx^2$, which becomes the launch kinetic energy $\frac{1}{2}mv_0^2$. Equating,

$$\tfrac{1}{2}kx^2 = \tfrac{1}{2}mv_0^2 \implies x = v_0\sqrt{\frac{m}{k}} = (4.91 \times 10^3)\sqrt{\frac{60}{1.2 \times 10^7}} = \boxed{11.0\ \mathrm{m}}.$$

Q5. Orbital force

By Newton's law of gravitation at the orbit,

$$F = \frac{GMm}{r^2} = \frac{(1.44 \times 10^{14})(60)}{(6.025 \times 10^6)^2} = \boxed{238\ \mathrm{N}},$$

in agreement with $F = mv_{\mathrm{orbit}}^2/r$.